Def^n: $G(s) \in \mathbb{R}^{p\times m}(s)$ ← real rational matrix

$\hat{G}(s) \in \mathbb{R}^{p\times m}(s)$ is called inverse of $G(s)$ if $G(s)\hat{G}(s) = \hat{G}(s)G(s) = I$.

$$G^{-1} \sim \begin{bmatrix} A - BD^{-1}C & -BD^{-1} \\ D^{-1}C & D^{-1} \end{bmatrix}$$

$\boxed{L12}$ 30/09/2020

Assume $G(s) \in \mathbb{R}^{p\times m}(s)$ is proper. Then we call a state-space model (A,B,C,D) such that $G(s) = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$ a realization of $G(s)$.

e.g. $\frac{2}{s+1}$ ✓ $\xleftrightarrow{\text{realization}}$ $\dot{x} = -x + u$, $y = 2x$ $\begin{bmatrix} -1 & 1 \\ 2 & 0 \end{bmatrix}$ ✓, $\begin{bmatrix} -1 & 2 \\ 1 & 0 \end{bmatrix}$ ✓

e.g. A realization (A,B,C,D) of $G(s)$ is called minimal realization of $G(s)$ if A has the smallest dimension possible.

e.g. $G(s) = \begin{bmatrix} \frac{1}{s+1} & \frac{2}{s+1} \end{bmatrix} \in \mathbb{R}^{1\times 2}(s)$

$\mathcal{C}_{AB} = [B_1 \;\; A_1B_1]$ $\mathcal{O}_{CA} = \begin{bmatrix} 1 & 2 \\ -1 & -2 \end{bmatrix}$ Unobservable

✗ $n=2$ ✓ $A_1 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$, $B_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $C_1 = [1 \;\; 2]$, $D_1 = 0$ $\quad [1 \;\; 2]\begin{bmatrix} s+1 & 0 \\ 0 & s+1 \end{bmatrix}^{-1}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = G(s)$

$n=1$ ✓ $A_2 = -1$, $B_2 = 1$, $C_2 = [1 \;\; 2]$, $D = 0$ $\quad [1 \;\; 2]\frac{1}{s+1} = G(s)$

A state space realization (A,B,C,D) of $G(s)$ is minimal $\Longleftrightarrow$ (A,B) is controllable & (C,A) is observable

Controller Canonical Form:

$$G(s) = \frac{\beta_{n-1}s^{n-1} + \beta_{n-2}s^{n-2} + \dots + \beta_1 s + \beta_0}{s^n + \alpha_{n-1}s^{n-1} + \dots + \alpha_0}$$

$\exists (\hat{A}, \hat{B}, \hat{C}, \hat{D})$ s.t. $G(s) = \begin{bmatrix} \hat{A} & \hat{B} \\ \hat{C} & \hat{D} \end{bmatrix}$, where

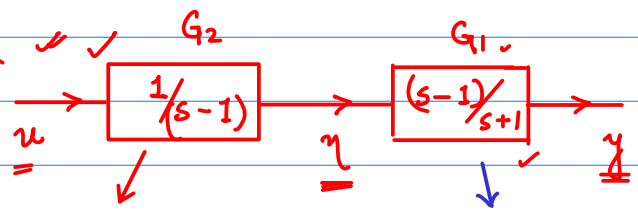
$$\hat{A} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & & & & \\ 0 & 0 & 0 & \cdots & 1 \\ -\alpha_0 & -\alpha_1 & -\alpha_2 & \cdots & -\alpha_{n-1} \end{bmatrix} \quad \hat{B} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} \quad \hat{C} = \begin{bmatrix} \beta_0 & \beta_1 & \cdots & \beta_{n-1} \end{bmatrix}$$

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$
$$x = T\hat{x}$$
$$\dot{\hat{x}} = T^{-1}AT\hat{x} + T^{-1}Bu \quad *$$
$$y = \underline{\underline{CT}}\hat{x} + \underline{\underline{D}}u$$

Observer canonical form: (C, A)

$$A = \begin{bmatrix} 0 & 0 & \cdots & -\alpha_0 \\ 1 & 0 & \cdots & -\alpha_1 \\ \vdots & \vdots & & \\ 0 & 0 & \cdots & -\alpha_{n-1} \end{bmatrix} \quad C = \begin{bmatrix} 0 & 0 & \cdots & 0 & 1 \end{bmatrix} \quad B = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_{n-1} \end{bmatrix}$$

e.g. $*$



$\dot{x}_2 = x_2 + u$

$\eta = x_2$

$\dot{x}_1 = -x_1 + \eta$

$y = -2x_1 + \eta$

$n = 2$

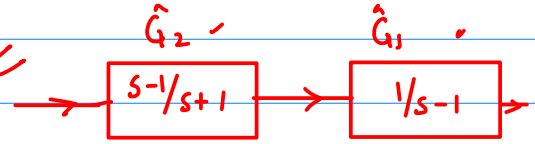
$\begin{bmatrix} A_1 & 0 \\ * & * \end{bmatrix}_{2\times 2}$

rank $\begin{bmatrix} \lambda I - A \\ C \end{bmatrix}$

1 → unobservable eigenvalue

controllable

$*$



uncont^r & observable

1

$G = 1/s+1$

$\boxed{\begin{aligned} \dot{x} &= -x + u \\ y &= x \end{aligned}} \Rightarrow 1$

Poles of system $\neq$ Eigenvalues of A

$\boxed{1}$ The eigenvalues of A (minimal rep.) are called the poles of the system.

e.g. $G(s) = \begin{bmatrix} 1/s+1 & 1/s+2 \\ 2/s+2 & 1/s+1 \end{bmatrix}$ $\sqrt{2}$

$-2, -2, -1, -1$

Stable

det $G(s)$

$K(s) = \begin{bmatrix} \frac{s+2}{s-\sqrt{2}} & \frac{s+1}{s-\sqrt{2}} \\ 0 & 1 \end{bmatrix}$

Unstable $\sqrt{2}$

$K(s)\,G(s)$

$-2, -1, -1$

$\mathbb{R}^{p\times m}[s] \leftarrow$ $p\times m$ matrix of polynomials with co-eff from $\mathbb{R}$

$\mathbb{R}^{p\times m}(s) \leftarrow$ $p\times m$ matrix of rational f^n with co-eff from $\mathbb{R}$.

Let $Q(s) \in \mathbb{R}^{p\times m}[s]$. Then normal rank of $Q(s)$ is the maximally possible rank of $Q(s)$ for atleast one $s \in \mathbb{C}$.

e.g. $Q(s) = \begin{bmatrix} s & 1 \\ s^2 & 1 \\ s & 1 \end{bmatrix}$.

$s = 1$, $\text{rank}(Q(1)) = 1$

$s = 2$, $\text{rank}(Q(2)) = 2 \leftarrow$ normal rank

$Q_2(s) = \begin{bmatrix} s+1 & 0 \\ 0 & s+2 \end{bmatrix}$ normal rank $(Q_2(s)) = 2$ $\quad$ $\text{nrank}(Q_2(s)) = 2$

$Q_3(s) = \begin{bmatrix} s+1 & s+1 \end{bmatrix}$ $\quad$ $\text{nrank}(Q_3(s)) = 1$

A complex number $z_0 \in \mathbb{C}$ is called invariant zero of the system realization if

$$\text{rank}\begin{bmatrix} A - z_0 I & B \\ C & D \end{bmatrix} < \text{nrank}\begin{bmatrix} A - sI & B \\ C & D \end{bmatrix}$$